



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE & TECHNOLOGY**

SCHOOL OF OPEN, DISTANCE AND eLEARNING

P.O. Box 62000, 00200

Nairobi, Kenya

E-mail: elearning@jkuat.ac.ke

ICS 2302 SOFTWARE ENGINEERING 1

JKUAT SODEL

©2017





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 2

Key Challenges Facing Software Engineering

Learning Outcomes

A study of this lesson should enable students to:

- Define the software engineering challenges
- Describe the software engineering methods



ICS 2302 Software Engineering 1

Software Engineering in 21st century faces 3 key challenges:-

- The legacy challenge – the majority of large software systems which are in use today were developed many years ago yet they perform critical business functions. The legacy challenge is the challenge of maintaining and updating this software in such a way that excessive costs are avoided and essential business services continue to be delivered.
- The heterogeneity challenge – increasingly systems are required to operate as distributed systems across networks that include different types of computers and devices with different kind of support systems. The heterogeneity challenge is the challenge of developing techniques to build dependable software which is flexible enough to cope with this heterogeneity.

JKUAT: Setting trends in higher Education, Research and Innovation



- The delivery challenge – the delivery challenge is the challenge of shortening delivery times for large and complex systems without compromising system quality.

Professional and Ethical Responsibility

Like other engineers, software engineers must accept that their job involves other responsibilities than simply the application of technical skills. Their work is carried out within a legal and social framework. It goes without saying that engineers should uphold normal standards of honesty and integrity. They should not use their skills and abilities to behave in a dishonest way or in a way that will bring disrepute to the software engineering profession. However there are areas where standards of acceptable behavior are not bounded notion of professional responsibility.



Some of these are:-

Software engineering ethics

- Software engineering involves wider responsibilities than simply the application of technical skills.
- Software engineers must behave in an honest and ethically responsible way if they are to be respected as professionals.
- Ethical behaviour is more than simply upholding the law but involves following a set of principles that are morally correct.

Issues of professional responsibility

- Confidentiality engineers – should normally respect the confidentiality of their employers/clients.



ICS 2302 Software Engineering 1

- Competence – engineers should not misrepresent their level of competence.
- Intellectual property rights – engineers should be aware of local laws governing the use of intellectual property such as patents, copyright, etc. They should be careful to ensure that the intellectual property of employers and clients is protected.
- Computer misuse – Software engineers should not use their technical skills to misuse other people's computers. Computer misuse ranges from relatively trivial (game playing on an employer's machine, say) to extremely serious (dissemination of viruses).

In this respect, professional societies and institutions have an important role to play. ACM and IEEE (Institute of Electrical



ICS 2302 Software Engineering 1

and Electronic Engineering) have cooperated to produce a joint code of ethics and professional practice. Its short form is below:-

Software engineers shall commit themselves to making the analysis, specification, design, development, testing and maintenance of software a beneficial and respected profession. In accordance with their commitment to high health safety and welfare of the public, software engineers shall adhere to the following eight principles:-

1. PUBLIC – software engineers shall act consistently with public interest.
2. CLIENT AND EMPLOYER – software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. PRODUCT – software engineers shall ensure that their



ICS 2302 Software Engineering 1

products and related modifications meet the highest professional standards possible.

4. JUDGEMENT – software engineers shall maintain integrity and independence in their profession at judgment.
5. MANAGEMENT – software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. PROFESSION – software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. COLLEGUES – software engineers shall be fair to and supportive to their colleagues.



ICS 2302 Software Engineering 1

8. SELF – software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.



Revision Questions

Example . What are the key challenges facing software engineering?

Solution:

The legacy challenge – Coping with increasing diversity

The heterogeneity challenge – build dependable software which is flexible enough

The delivery challenge – demands for reduced delivery times



EXERCISE 1.  What is software engineering methods?



References and Additional Reading Materials

1. Hans van Vliet (2005). Software Engineering: Principles and Practice (2nd ed.). John Wiley & Sons, Inc. ISBN: 0-471-97508-7
2. Ian Sommerville, Pete Sawyer(2005). Requirements Engineering: A Good Practice Guide. ISBN: 0-471-97444-7. John Wiley & Sons, Inc
3. Gerald Kotonya, Ian Sommerville (1998). Requirements Engineering: Processes and Techniques. John Wiley & Sons, Inc. ISBN: 0-471-97208-8